HEART DISEASE PREDICTION REPORT

Explainable Deep Learning Diagnosis Platform (CNN-LSTM-Attention + Grad-CAM)

Patient Index	#8	**True Label (Ground Truth)**	Heart Disease
Model Prediction Probability	82.90%	**Analysis Date**	2026-06-23

Model Recommendation: HEART DISEASE PREDICTED

Patient Clinical Profile

Age	35 years	**Sex**	Female
Chest Pain Type (CP)	Asymptomatic	**Resting Blood Pressure**	126 mm Hg
Serum Cholesterol	282 mg/dl	**Fasting Blood Sugar (FBS)**	Normal (<=120 mg/dl)
Resting ECG	Left Ventricular Hypertrophy	**Max Heart Rate (Thalach)**	156 bpm
Exercise Induced Angina	Yes	**ST Depression (Oldpeak)**	0.0
ST Slope	Upsloping	**Vessels via Fluoroscopy (CA)**	0
Thalassemia (Thal)	Reversible Defect	**Predicted Diagnosis Probability**	**82.90%**

Primary Diagnostic Drivers (AI Focus)

1. CP

Patient Value: 4.0 | **Ideal Range:** 1, 2, or 3 (Angina/Non-anginal pain)

Interpretation: *Asymptomatic chest pain (type 4). This silent presentation represents a high risk factor as blockages are painless.*

2. CA

Patient Value: 0.0 | **Ideal Range:** 0 vessels colored

Interpretation: *0 major vessels colored (clear arterial paths).*

3. RESTECG

Patient Value: 2.0 | **Ideal Range:** 0 (Normal)

Interpretation: *Abnormal ECG resting results (type 2), suggesting hypertrophy or ST-T wave anomalies.*

4. THAL

Patient Value: 7.0 | **Ideal Range:** 3 (Normal)

Interpretation: *Reversible defect thalassemia observed (indicating active myocardial ischemia under stress).*

Secondary Clinical Abnormalities (Outside Top AI Focus)

1. CHOL

Patient Value: 282.0 | **Ideal Range:** <200 mg/dl

Interpretation: *Measured at 282.0 mg/dl. High cholesterol level, indicating severe plaque build-up risk in coronary arteries.*

2. THALACH

Patient Value: 156.0 | **Ideal Range:** >157 bpm (85% of max predicted 185 bpm)

Interpretation: *Measured at 156.0 bpm (only 84.3% of age-predicted max). This chronotropic incompetence suggests restricted cardiac output.*

3. EXANG

Patient Value: 1.0 | **Ideal Range:** 0 (No)

Interpretation: *Angina (chest pain) induced during exercise stress, indicative of myocardial hypoxia.*

Explainability Visualization (Attention vs. Grad-CAM)

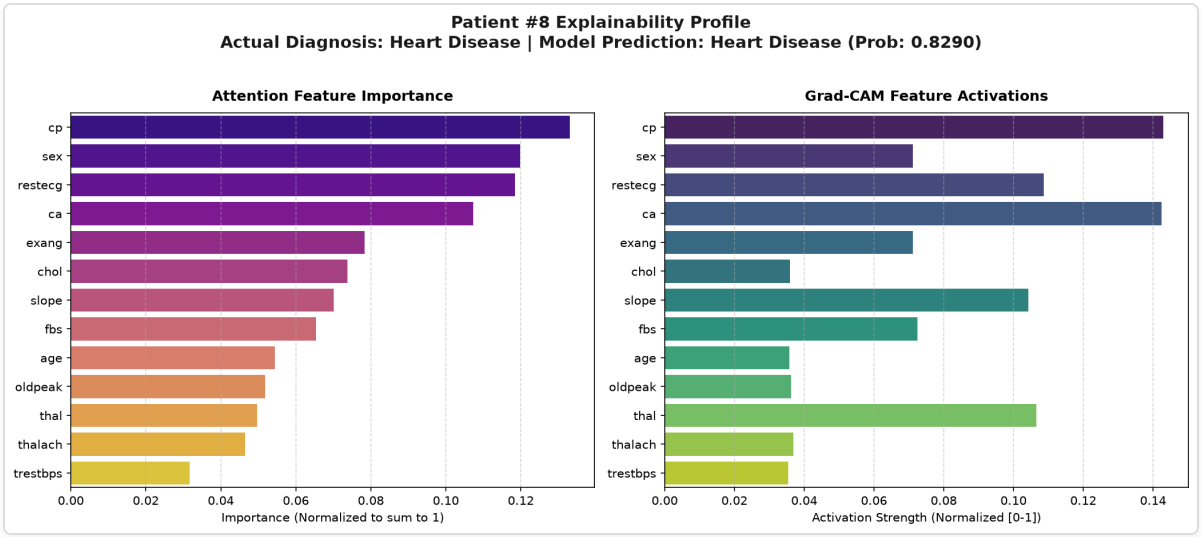


Figure 1: Dual-Perspective feature attribution showing global attention weight and patient-specific Grad-CAM contribution.

2D Gramian Angular Field (GASF) Heatmap

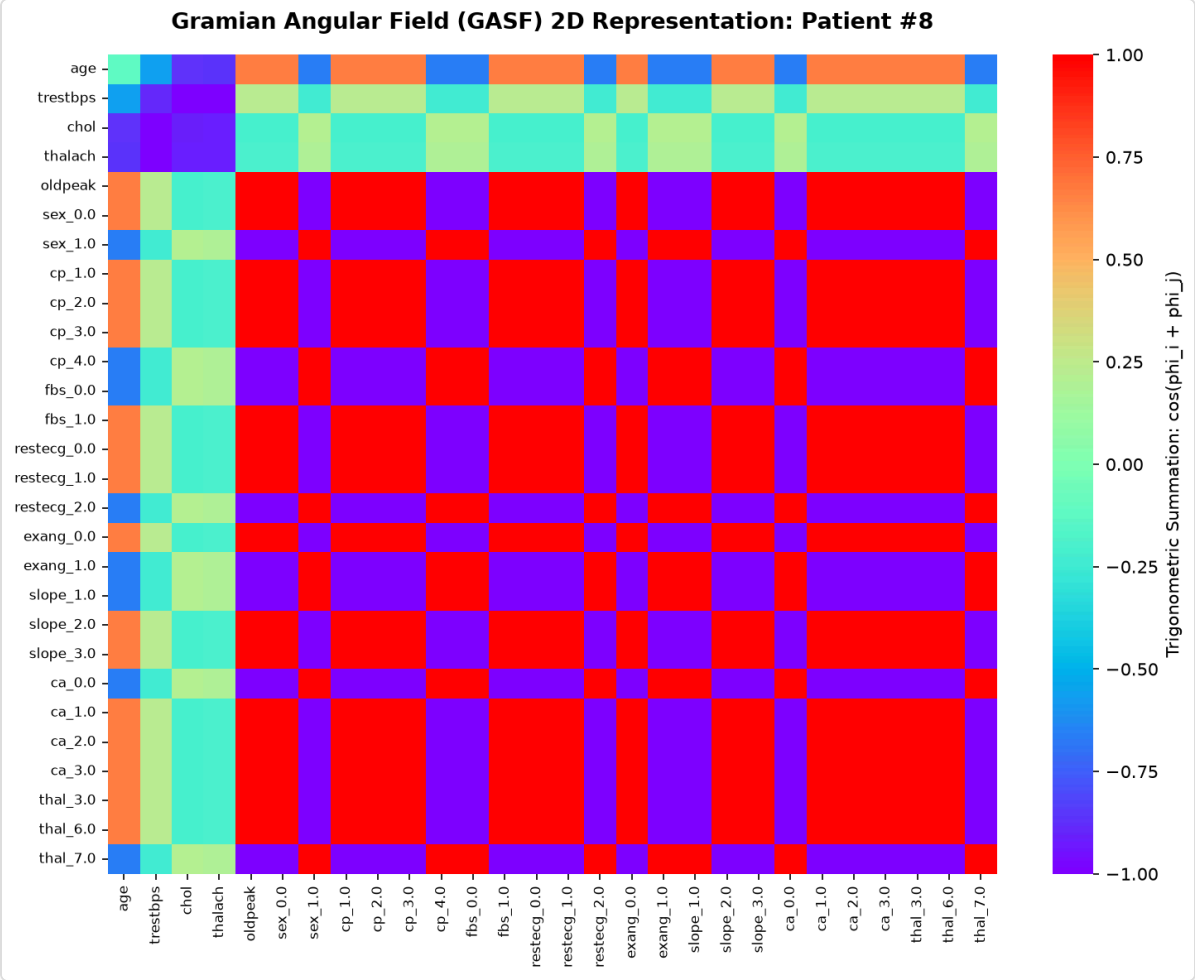


Figure 2: 2D feature-feature interaction texture mapping derived via Gramian Angular Summation Field conversion.